A. Flag

time limit per test: 2 seconds
memory limit per test: 64 megabytes

According to a new ISO standard, a flag of every country should have a chequered field $n \times m$, each square should be of one of 10 colours, and the flag should be «striped»: each horizontal row of the flag should contain squares of the same colour, and the colours of adjacent horizontal rows should be different. Berland's government asked you to find out whether their flag meets the new ISO standard.

Input

The first line of the input contains numbers n and m ($1 \le n, m \le 100$), n — the amount of rows, m — the amount of columns on the flag of Berland. Then there follows the description of the flag: each of the following n lines contain m characters. Each character is a digit between 0 and 9, and stands for the colour of the corresponding square.

Output

Output YES, if the flag meets the new ISO standard, and NO otherwise.

Examples

input Copy

```
3 3
000
111
222
```

output Copy

```
YES
```

input Copy

```
3 3
000
000
111
```

output Copy

```
NO
```

input Copy

```
3 3
000
111
002
```

output Copy

```
NO
```

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 16 (Div. 2 Only)

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
370349180	Apr/09/2026 08:42	Accepted

→ Problem tags

implementation *800

No tag edit access

→ Contest materials

- Announcement
- Tutorial

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